

MELEK BOUABID

Computer Engineering Student — Data Science & Artificial Intelligence

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Profile

Computer Engineering student, specialized in Data Science and Artificial Intelligence at Tek-Up. Experience in Machine Learning, NLP and RAG systems, with projects covering data preparation, model training and deployment.

Technical Skills

- **Programming:** Python, SQL, JavaScript
- **Data Science & ML:** Pandas, NumPy, Scikit-learn, XGBoost, Random Forest, feature engineering
- **Deep Learning & AI:** TensorFlow, PyTorch, NLP, LangChain, RAG, vector databases
- **Data Engineering:** PySpark, PostgreSQL, MySQL
- **Deployment & Tools:** FastAPI, Docker, MLflow, Git
- **Web Development:** React, Django

Education

Computer Engineering Degree — Data Science & AI	2024 – 2027
Tek-Up, Tunisia — 5 th year (in progress)	
B.Sc. in Environmental Physics-Chemistry	2019 – 2023
Higher Institute of Environmental Sciences and Technologies of Borj Cédria, Tunisia	

Professional Experience

Data Science / AI Intern — STIET-Philips	2026 (2 months)
• Contributed to a smart CMMS POC (Laravel 12) for tracking medical installations (MRI and CathLab rooms), structuring operational, technical, financial and documentary data across the installation lifecycle. • Implemented a conversational assistant based on a documentary RAG architecture (document indexing and chunking, relevant passage retrieval, answer generation via the Gemini API) for administrators and managers. • Designed decision-support scoring models: equipment risk score (warranty, age, incidents, complaints, maintenance history), team performance score, and statistical prediction of installation duration from archived data. • Took part in designing an interactive installation-readiness graph (equipment, components, safety elements, documents) with status indicators, and in setting up caching and re-indexing mechanisms.	
Mobile Development Intern — Advanced Technologies Software	2025 (1 month)
• Built a Flutter application to scan Ray-Ban products' barcodes and QR codes and check stock availability.	

Projects

CSIS — Criminal Study Intelligent System	2025
• Social risk detection among young people: ML (XGBoost, Random Forest) + NLP on text data. Legal assistant on a RAG architecture (LangChain, vector database) grounded in Tunisian law, presented via Streamlit & Power BI.	
Riot Games Analytics Platform	2026
• Built classification and regression models to predict match outcomes and player rank. • Data pipeline with the Riot Games API, experiment tracking with MLflow, deployed with FastAPI and React.	
Generator Data Crime	
• Generated synthetic criminology datasets (JSON/CSV) with realistic statistical correlations, used to test data processing pipelines. <i>Technologies: React, Vite.</i>	
The Room — Escape Game Booking Website	
• Public online booking website with an admin dashboard, real-time notifications and data export. <i>Technologies: React, Express, PostgreSQL, Docker, SSE.</i>	
E-Bike Rental Platform	
• Peer-to-peer electric bike rental marketplace with authentication and online payments. <i>Technologies: Django REST Framework, React, TypeScript, JWT, Stripe.</i>	

Languages

Arabic: Native | French: Fluent | English: Fluent